

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
• MCP pipelines (Claude, GPT-4o, Gemini) with tool calling; PDF gen, S3, email parsing; Playwright tests. • Owned 7-deployment Node.js backend: PostgreSQL, Redis, Auth0 RBAC, Temporal.io, Docker, Terraform, Sentry.		
Graduate Research Assistant	NC State University	April 2025 – Present
• Cloud-based genomic platform on Google Cloud Run with JBrowse 2; 100+ researchers, 500GB datasets worldwide. • Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
• Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved processing accuracy by 25%. • Automated deployments on AWS and GCP; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
• Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. • Built Kafka + Spark + Hadoop pipelines for batch and real-time ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
• Master of Science in Computer Science • Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
• Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

LLM Eval Harness	Independent Project	2025
• Vendor-neutral benchmark comparing Claude, GPT-4o, Gemini across 2,000 examples and 3 task types in Python. • Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		
VectorLift	Semantic Search Engine	2026
• Semantic search in Python on MS MARCO: BM25, dense bi-encoder (in-batch negatives), cross-encoder re-ranker. • Scored NDCG@10, MRR@10, MAP; FastAPI serving with per-stage latency; Kafka-driven Qdrant corpus indexing.		
AlgoPulse	Agentic AI Platform	2025
• Agentic AI orchestration on Temporal.io and Kafka; 2,000 async inference req/sec without state loss. • Replaced API polling with Supabase Edge Functions streaming AI reasoning loops; no state loss under load.		
Glance AI	Privacy-First RAG	2025
• Per-tenant RAG for healthcare with vector isolation preventing cross-tenant data leakage and hallucinations. • Deterministic retrieval architecture for regulated environments; factual LLM outputs, no context bleed.		

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **AI/ML & Protocols:** Claude/GPT-4o/Gemini APIs, MCP, OpenAI API, function calling, JSON-RPC, RAG, LLM evaluation
- **Backend & Cloud:** FastAPI, Node.js, NestJS, Temporal.io, Kafka, Docker, Kubernetes, GitHub Actions, AWS, GCP
- **Data:** PostgreSQL, Redis, Elasticsearch, Qdrant, DuckDB, Spark

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog